

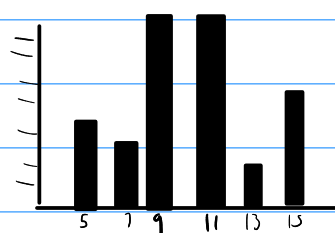
0.6 Statistical reasoning in everyday life

Often, an off-the-top-of-the-head estimate giving a nice round number may cement itself in our collective consciousness

What are descriptive statistics?

Researchers may use descriptive statistics to measure and describe characteristics of the group under study. Example: a histogram

When looking at histograms, be wary of the range of the y-axis which can easily be manipulated to push a message



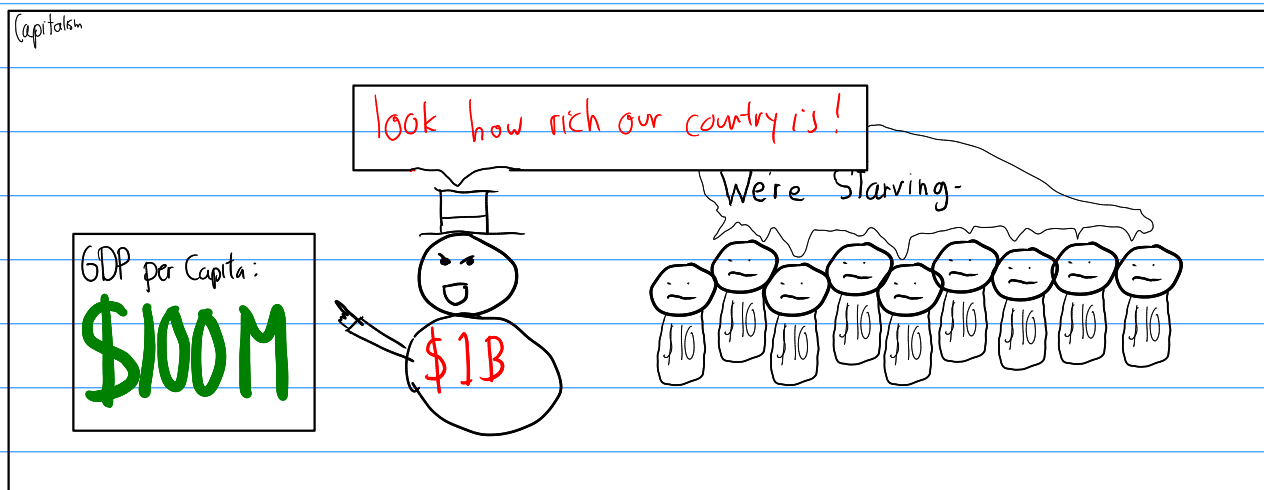
How can we summarize data?

Researchers summarize data using a measure of central tendency

- a mode: most frequent value
- mean: arithmetic average
- median: (think medium) midpoint of data distribution

Percentage rank is the percentage of scores less than some score

Data can be skewed by a few extreme examples



Personal Example

Esp. because I use Linux, I have a very large amount of very small programs on my laptop. Most programs on my laptop are very small. But because of a small number of large programs like Firefox, etc. the mean app size is still large.

How can we measure variation?

It can be very useful to know the amount of variation in data

The range of scores only gives a crude approximation of variation

$$\sigma = \sqrt{\frac{\sum_{i=0}^{|\text{scores}|} (\text{score}_i - \overline{\text{score}})^2}{|\text{scores}| - 1}}$$

A more useful tool is Standard Deviation. It is a very useful measure of variation.

Large numbers of data often follow a bell-shaped distribution called the normal curve

$$\frac{|\{\text{case} \mid \mu - \sigma \leq \text{case} \leq \mu + \sigma, \text{case} \in \{\text{cases}\}\}|}{|\{\text{cases}\}|} \approx 0.68$$

$$\frac{|\{\text{case} \mid \mu - 2\sigma \leq \text{case} \leq \mu + 2\sigma, \text{case} \in \{\text{cases}\}\}|}{|\{\text{cases}\}|} \approx 0.95$$

What are inferential statistics?

Differences between those in the control group and the experimental group may appear as a result of random fluctuations.

Using inferential statistics helps to determine the generalizability of results.

It is about drawing conclusions that go further than the raw data

When are observed differences reliable?

When deciding if to infer Δp from $\Delta \hat{p}$:

- Representative samples are better than biased samples
- Bigger samples are better
- More estimates are better; conduct >1 study, combine estimates, using meta-analysis

p - population proportions
 $\hat{p}$ - sample proportions

Researchers use statistical testing to estimate $P(\text{result happening randomly})$

They assume the null hypothesis, that no difference exists between groups

Using statistics, they observe if $P(\Delta\hat{p})$ is so rare ($< .05$) it's unlikely to fit the null hypothesis. If so, they say that the result is statistically significant.

$$P(\Delta\hat{p} | H_0) < .05 \Rightarrow \text{Statistical Significance}$$

Statistical significance $\neq$ practical significance
Esp. very large sample $\Rightarrow$ result can be Stat. Sig., but have very low effect size.

Researchers also use the confidence interval: {range}, $P(\bar{x} \in \text{range})$ is likely

Ex. $\bar{x}_{\text{score}} = 80 \nRightarrow \mu_{\text{score}} = 80$
StatAn gives two values, for ex. 70 & 90
 $CJ = \{\text{range}\}, \forall x \in \text{range} (70 \leq x \leq 90), P(\mu_{\text{score}} \in \text{range}) = \text{likely}$

Summary

Descriptive Stats

- to measure / describe
- often uses histograms
- incl. $\begin{cases} \text{Central Tendency} \\ \text{Percentile rank} \\ \text{Measures of variation} \end{cases}$

Inferential Stats

- infer
- predictions
- incl. ways of determining reliability & significance of $\Delta\hat{p}$

Central Tendency

- mean = $\overline{\text{score}}$
 - mode = $\{x | x \in \{\text{scores}\}, \exists k \neq n \text{ s.t. } \exists c \in \{\text{scores}\}, \forall a \geq n (\neg \exists b \in \{\text{scores}\})\}$
 - median = x , $\begin{cases} |\{\text{scores}\}| \equiv 1 \Rightarrow x \in \{\text{scores}\} \wedge (|A|=|B|, A \cup B = \{\text{scores}\}, \forall n \in A (n \leq x), \forall n \in B (n \geq x)) \\ |\{\text{scores}\}| \equiv 2 \Rightarrow x = \frac{a+b}{2}, a = \max(A), b = \min(B), |A|=|B|, \forall n \in A (n \leq b), \forall n \in B (n \geq a) \end{cases}$
 - percentile score (n) = $\frac{|\{\text{score} | \text{score} \leq n, \text{score} \in \{\text{scores}\}\}|}{|\{\text{scores}\}|}$
- $\forall y \in \{\text{scores}\} (y \neq x \Rightarrow |\overline{\text{score}} - y| > |\overline{\text{score}} - x|)$

Measures of variation

- range = $\max(\{\text{scores}\}) - \min(\{\text{scores}\})$
crude, inaccurate measure of variation
- standard deviation = $\sqrt{\frac{\sum (\text{deviation}^2)}{|\{\text{scores}\}| - 1}}$
(how much scores vary around mean)
scores often make normal curve

Generalizing Data:

- Reliable differences are based on samples that $\left\{ \begin{array}{l} \text{represent larger population} \\ \text{demonstrate low variability} \\ \text{consist of many cases} \end{array} \right.$
- $P(\Delta\hat{p} | H_0) < .05 \Rightarrow$ statistical significance
- CI = {range}, $\forall x \in \text{range} (a \leq x \leq b)$, a,b given by statistic
 $P(\mu_{\text{scores}} \in \text{CI}) = \text{likely}$

MCQs: 1d, 2a, 3d, 4a, 5b, 6d, 7c